STPM 2017 PHYSICS 960/2

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $N_A = 6.02 \times 10^{23}$ mol^{-1}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}. Q1: the answer is C (the force on a unit positive charge).

Section A [15 marks]

1. The electric field strength is the

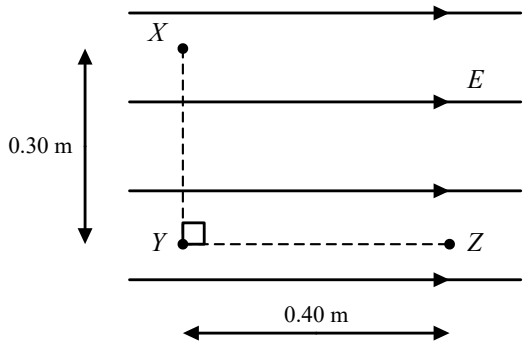
A work done to bring a proton at a point
B electrostatic force that acts on a proton placed at a point
C electrostatic force that acts on a unit of positive charge at a point
D work done to bring a unit of positive charge from infinity to a point

Answer: C

Electric field strength $E = \frac{F}{q}$: the electrostatic force acting on a unit positive charge at that point (C)

Explanation: D defines electric potential; A and B do not refer to a unit charge.

2. Three points, X, Y and Z, in a uniform electric field E of magnitude 200 V m^{-1} are shown in the diagram below.



What is the work done to move an electron from X to Z?

A 1.3×10^{-17} J B 1.6×10^{-17} J C 2.2×10^{-17} J D 3.2×10^{-17} J

Answer: A

X to Y is perpendicular to the field: no work. Y to Z is 0.40 m along the field.

The electron is pushed against the field, so work must be done: $W = eEd = (1.60 \times 10^{-19})(200)(0.40) = 1.3 \times 10^{-17}$ J

Explanation: Only the displacement along the field matters.

3. Two capacitors which have the same capacitance can be connected in series or parallel. Which statement is true about the charge stored for the same d.c. voltage supply?

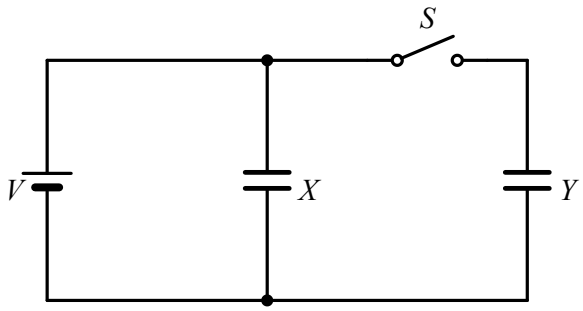
A The series connection stores an amount of charge twice that of the parallel connection.
B The parallel connection stores an amount of charge twice that of the series connection.
C The series connection stores an amount of charge four times that of the parallel connection.
D The parallel connection stores an amount of charge four times that of the series connection.

Answer: D

Parallel: $2C$; series: $\frac{C}{2}$. With the same V, $Q = CV$: parallel stores four times the charge of series

Explanation: The ratio of capacitances is $2C : C/2 = 4 : 1$.

4. Two identical capacitors, X and Y, which are connected in parallel to a battery are shown in the diagram below.



When switch S is opened, the potential difference across X is V, the charge stored is Q and Y is uncharged. When switch S is closed, what is the potential difference across Y and the charge stored in Y?

	Potential difference	*Charge stored*
A	$\frac{1}{2}V$	Q
B	$\frac{1}{2}V$	$2Q$
C	V	Q
D	V	$2Q$

Answer: C

With S closed, Y is connected directly across the battery, just like X: p.d. V and charge Q (C).

Explanation: The battery keeps the p.d. at V; the capacitors are identical.

5. A conductor of cross-sectional area 5.0×10^{-6} m^2 carries a current of 6.0 A. The conductivity of the conductor at room temperature is 3.13×10^{7} Ω^{-1} m^{-1}. What is the electric field strength in the conductor?

A 3.8×10^{-2} V m^{-1}
B 1.9×10^{-7} V m^{-1}
C 9.6×10^{-13} V m^{-1}
D 2.7×10^{-14} V m^{-1}

Answer: A

$$J = \frac{6.0}{5.0 \times 10^{-6}} = 1.2 \times 10^{6}\ \text{A m}^{-2};\ E = \frac{J}{\sigma} = \frac{1.2 \times 10^{6}}{3.13 \times 10^{7}} = 3.8 \times 10^{-2}\ \text{V m}^{-1}$$

Explanation: $J = \sigma E$.

6. A copper wire of diameter 1.5 mm has an electric resistance of 4.0 Ω. If the resistivity of the copper is 1.7×10^{-8} Ω m, the length of the copper wire is

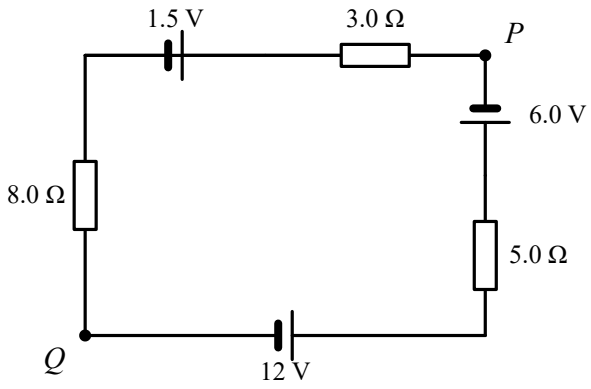
A 2.4 m B 26 m C 420 m D 1700 m

Answer: C

$$l = \frac{RA}{\rho} = \frac{(4.0)\pi(0.75 \times 10^{-3})^2}{1.7 \times 10^{-8}} = 416\ \text{m} \approx 420\ \text{m}$$

Explanation: Use the radius, half of 1.5 mm.

7. Three resistors of resistance 3.0 Ω, 5.0 Ω and 8.0 Ω, and three batteries of e.m.f. 1.5 V, 6.0 V and 12 V are connected in series as shown in the circuit below.



The electric potential at point P with respect to point Q is

A −15 V B −4.6 V C 4.6 V D 15 V

Answer: B

Taking the polarities as drawn, the 1.5 V and 6.0 V cells oppose the 12 V cell: net e.m.f. = 12 − 1.5 − 6.0 = 4.5 V.

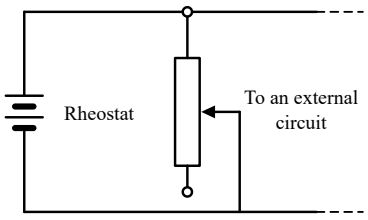
$$I = \frac{4.5}{3.0 + 5.0 + 8.0} = 0.281\ \text{A}$$

From Q to P through the 8.0 Ω, 1.5 V cell and 3.0 Ω: $V_P - V_Q = -(0.281)(11.0) - 1.5 = -4.6$ V

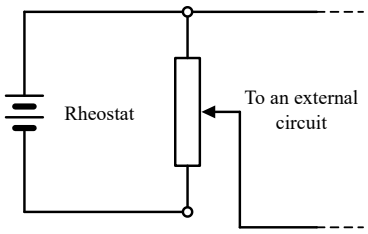
Explanation: Check through the other branch: $-12 + (0.281)(5.0) + 6.0 = -4.6$ V.

8. A rheostat may be used as a potential divider. Which circuit shows a rheostat being used to provide a fraction of voltage of the power supply to an external circuit?

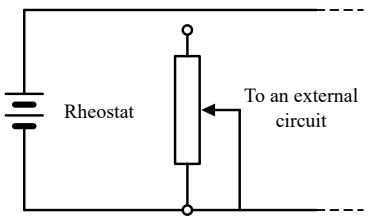
A



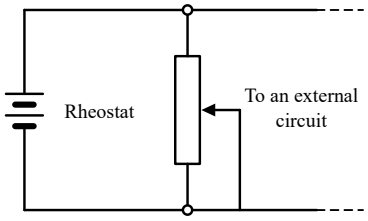
B



C



D

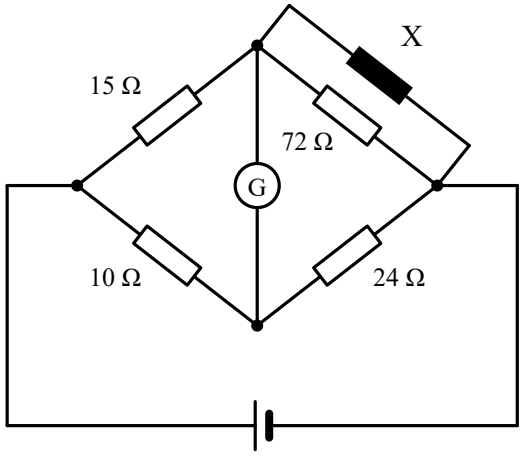


Answer: B

As a potential divider, the supply is connected across the whole rheostat and the output is taken between one end and the slider (B).

Explanation: The output can then be varied from zero to the full supply voltage.

9. A resistor X which is connected in parallel with a resistor of 72 Ω in a Wheatstone bridge circuit is shown in the diagram below.



The bridge will be in equilibrium if X has a resistance of

A 23 Ω B 36 Ω C 48 Ω D 72 Ω

Answer: D

$$\text{Balance: } \frac{15}{10} = \frac{R}{24} \Longrightarrow R = 36\ \Omega, \text{ where } R = X \parallel 72\ \Omega$$

$$\frac{1}{X} + \frac{1}{72} = \frac{1}{36} \Longrightarrow X = 72\ \Omega$$

Explanation: Two equal resistors in parallel give half the resistance.

10. Which source **cannot** produce a magnetic field?

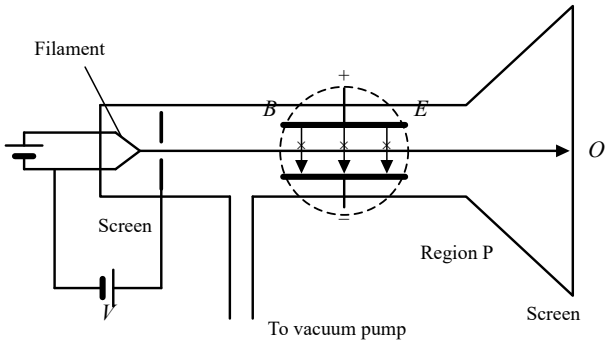
A The Earth
B A solenoid
C A permanent magnet
D A conductor which carries current

Answer: B

A solenoid produces a magnetic field only when it carries a current; on its own it is just a coil of wire (B).

Explanation: The Earth, a magnet and a current-carrying conductor always have fields.

11. In Thomson's experiment, a beam of electrons is allowed to pass through region *P* which consists of electric field *E* and magnetic field *B*, so that the electrons hit the screen at *O* as shown in the diagram below.



The function of region *P* is to

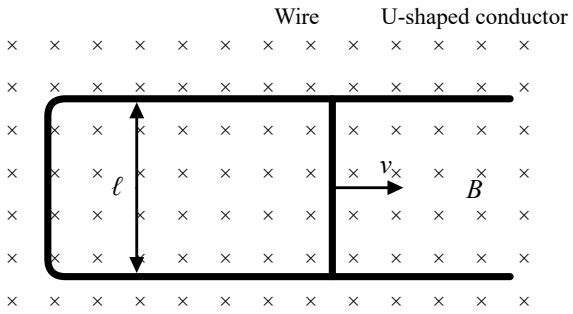
A accelerate the electrons
B decelerate the electrons
C focus the electrons onto the screen
D allow electrons of a certain speed to move through the region

Answer: D

Crossed fields select electrons with $eE = evB$, i.e. $v = \frac{E}{B}$: only those pass undeflected to *O* (D)

Explanation: This is a velocity selector.

12. A wire moved by an external force at a constant velocity v along a U-shaped conductor of width ℓ is shown in the diagram below.



A uniform magnetic field B is directed perpendicularly into the plane of the U-shaped conductor. If the resistance of the wire is R and the resistance of the U-shaped conductor is negligible, the external force applied on the wire is

A $B\ell v$ B $\frac{B\ell v}{R}$ C $B^2\ell^2 v$ D $\frac{B^2\ell^2 v}{R}$

Answer: D

$\varepsilon = B\ell v$, $I = \frac{B\ell v}{R}$; the force needed balances $BI\ell = \frac{B^2\ell^2 v}{R}$

Explanation: At constant velocity the resultant force is zero.

13. A current which changes at the rate of 1.5 A s^{-1} in coil P induces an e.m.f. of 0.48 V in coil Q. Which statement is true of the electromagnetic induction in coils P and Q?

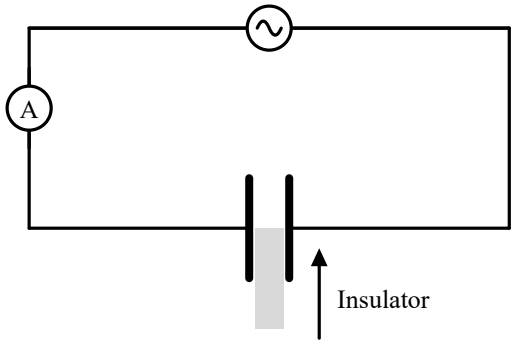
A Coils P and Q have a mutual inductance of 0.32 H.
B Coils P and Q have a mutual inductance of 0.72 H.
C An e.m.f. of 0.48 V is induced in coil Q because the magnetic flux through coil P is changing.
D An e.m.f. of 0.48 V must be induced in coil P because the magnetic flux through coil P is changing.

Answer: A

$$M = \frac{\varepsilon_Q}{\mathrm{d}I_P/\mathrm{d}t} = \frac{0.48}{1.5} = 0.32 \text{ H}$$

Explanation: The e.m.f. in Q is due to the changing flux from P linking Q.

14. An alternating voltage which is supplied to a pure capacitor in series with an a.c. ammeter is shown in the circuit below.



When an insulator is inserted into the capacitor, the ammeter reading

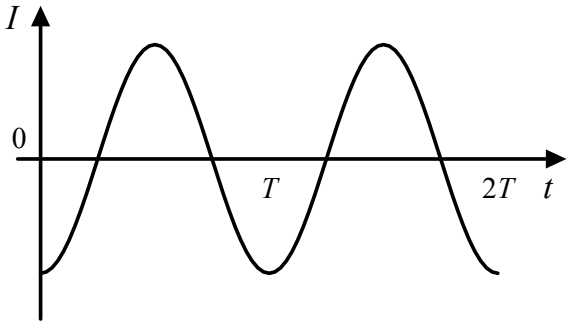
A increases because the reactance of the capacitor decreases
B decreases because the resistance of the insulator is very high
C remains unchanged because the time constant of the circuit decreases
D remains unchanged because the reactance of the capacitor is unchanged

Answer: A

The insulator increases C, so $X_C = \frac{1}{2\pi fC}$ decreases and the current increases (A)

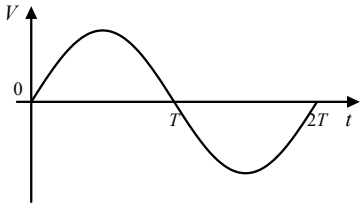
Explanation: A dielectric raises the capacitance.

15. The variation of current I with time t for a pure inductor is shown in the graph below.

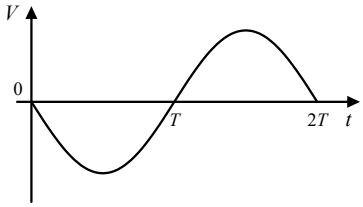


Which graph represents the variation of voltage V with time t for the inductor?

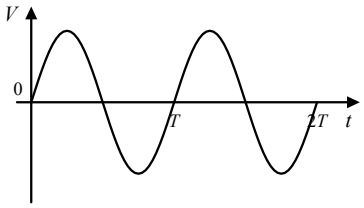
A



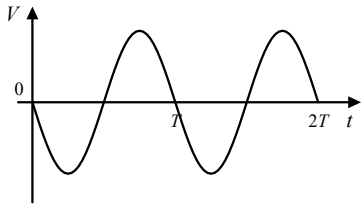
B



C



D



Answer: C

$V = L\frac{dI}{dt}$: I starts at its minimum, so $dI/dt = 0$ and then positive; V starts at zero and rises, with the same period T (C)

Explanation: The voltage leads the current by a quarter of a period.

Section B [15 marks]

16. A current of 15 A flows through a copper cube of side 10 cm. Assume each copper atom contributes one free electron.

[Density of copper is 8940 kg m^{-3} and molar mass of copper is 0.0635 kg mol^{-1}.]

(a) Calculate the number of free electrons in the copper cube. [4]

Answer

$n = \frac{\rho N_A}{M}$ 1

$n = \frac{(8940)(6.02 \times 10^{23})}{0.0635} = 8.48 \times 10^{28}\ m^{-3}$ 1

Volume of cube $= (0.10)^3 = 1.0 \times 10^{-3}\ m^3$ 1

$N = nV = 8.48 \times 10^{25}$ 1

4

Explanation: One free electron per atom.

(b)

(i) State the main factor that causes the drift velocity of the free electrons. [1]

Answer

The electric field (potential difference) applied across the conductor 1

1

Explanation: It accelerates the electrons between collisions.

(ii) Determine the drift velocity of the free electrons. [2]

Answer

$v_d = \frac{I}{nAe} = \frac{15}{(8.48 \times 10^{28})(0.010)(1.60 \times 10^{-19})}$ 1

$v_d = 1.1 \times 10^{-7}$ m s^{-1} 1

2

Total for question **7**

Explanation: The cross-section is $(0.10 \text{ m})^2 = 0.010 \text{ m}^2$.

17. An electromagnetic coil of resistance 300 Ω and inductance 5.0 H is connected to an alternating voltage supply of amplitude 240 V and frequency 60 Hz.

(a) Calculate the impedance of the coil. [3]

Answer

$X_L = 2\pi fL = 2\pi(60)(5.0) = 1885\ \Omega$ 1

$Z = \sqrt{(300^2 + 1885^2)}$ 1

$Z = 1.91 \times 10^3\ \Omega$ 1

3

Explanation: The coil's resistance and reactance act in series.

(b) Calculate the r.m.s. current. [3]

Answer

$V_{rms} = \frac{240}{\sqrt{2}} = 170$ V 1

$I_{rms} = \frac{170}{1909}$ 1

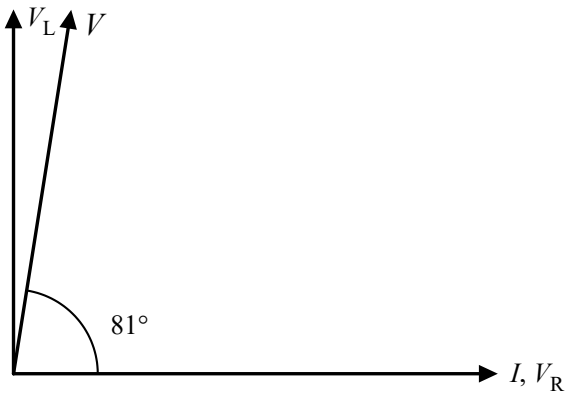
$I_{rms} = 0.089$ A 1

3

Explanation: 240 V is the amplitude (peak) voltage.

(c) Sketch the phasor diagram of the circuit. [2]

Answer



V_R in phase with I, V_L 90° ahead of I	1
Resultant V leading I by $\tan^{-1}(1885/300) = 81°$	1
	2
Total for question	**8**

Explanation: The coil is mainly inductive.

Section C [30 marks]

18. (a) Describe the changes in electrostatic quantities of a parallel plate capacitor when a dielectric material is inserted between the plates. [3]

Answer

The dielectric is polarised, so the capacitance increases by the factor ε_r	1
With the battery connected: V stays the same, so the charge (and energy) stored increase	1
If the capacitor is isolated: Q stays the same, while the field E and the p.d. V decrease by ε_r	1
	3

Explanation: In both cases $C = \varepsilon_r C_0$.

(b) Two identical air gapped parallel plate capacitors, A and B, each has a capacitance of 20.0 μF. A material with dielectric constant of 3.55 is inserted into the air gap of capacitor B. The two capacitors are then connected in series to a battery of 12.0 V. Determine

(i) the effective capacitance of the circuit, [3]

Answer

$C_B = 3.55 \times 20.0 = 71.0$ μF — 1

Series: $C = \dfrac{(20.0)(71.0)}{20.0 + 71.0}$ — 1

$C = 15.6$ μF — 1

3

Explanation: Series capacitance is less than either.

(ii) the energy stored in each capacitor in the circuit. [4]

Answer

Charge on each (series): $Q = CV = (15.6 \times 10^{-6})(12.0) = 1.87 \times 10^{-4}$ C — 1

$U_A = \dfrac{Q^2}{2C_A} = \dfrac{(1.87 \times 10^{-4})^2}{2(20.0 \times 10^{-6})}$ — 1

$U_A = 8.8 \times 10^{-4}$ J — 1

$U_B = \dfrac{(1.87 \times 10^{-4})^2}{2(71.0 \times 10^{-6})} = 2.5 \times 10^{-4}$ J — 1

4

Explanation: In series the smaller capacitor stores more energy.

(c) A capacitor of capacitance 47.0 μF is fully charged by a battery of 6.0 V. The capacitor is then connected to a discharging circuit of a camera flash containing a resistance of 200 kΩ.

(i) Determine the time constant. [2]

Answer

$\tau = RC = (200 \times 10^3)(47.0 \times 10^{-6})$ — 1

$\tau = 9.4$ s — 1

2

Explanation: Time for the charge to fall to 37%.

(ii) Calculate the amount of charge remaining in the capacitor after 10.0 s. [2]

Answer

$Q_0 = (47.0 \times 10^{-6})(6.0) = 2.82 \times 10^{-4}$ C; $Q = Q_0 e^{-t/\tau} = 2.82 \times 10^{-4}\ e^{-10.0/9.4}$ 1

$Q = 9.7 \times 10^{-5}$ C 1

2

Explanation: About one third of the charge remains.

(iii) State the effect on the power of the camera flash if a smaller value of resistance is used. [1]

Answer

The power increases (the energy is released in a shorter time) 1

1

Total for question **15**

Explanation: A smaller R gives a smaller time constant.

19. (a) State Ampere's law. [1]

Answer

The line integral of **B** around any closed path equals μ_0 times the current enclosed: $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$ 1

1

Explanation: The path is called an Amperian loop.

(b) Using Ampere's law, derive the magnetic field B at the distance r from an infinitely long straight wire which carries current I. [3]

Answer

Choose a circle of radius r centred on the wire; by symmetry **B** is tangential with constant magnitude 1

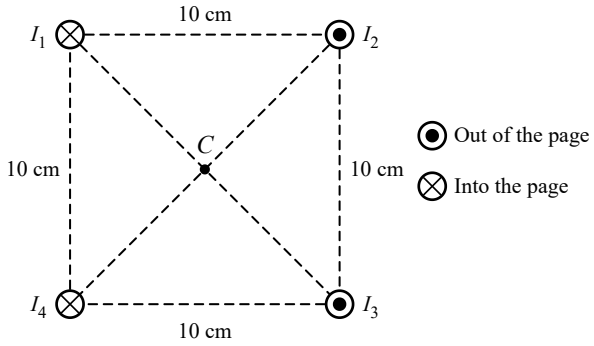
$\oint \mathbf{B} \cdot d\mathbf{l} = B(2\pi r) = \mu_0 I$ 1

$B = \frac{\mu_0 I}{2\pi r}$ 1

3

Explanation: The direction follows the right-hand grip rule.

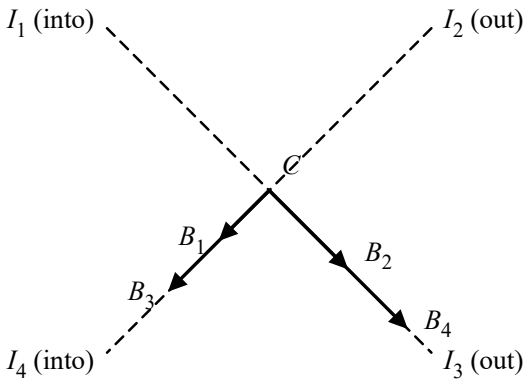
(c) A cross-sectional view of four cables which carry current is shown in the diagram below.



The currents I_1, I_2, I_3 and I_4 are 16 A, 25 A, 32 A and 44 A respectively.

(i) Draw the magnetic fields at point C due to the respective currents. [2]

Answer



Each field perpendicular to the line joining C to that wire	1
Correct senses: B_1, B_3 towards I_4; B_2, B_4 towards I_3	1
	2

Explanation: Right-hand grip rule for each current.

(ii) Calculate the magnitude and direction of the resultant magnetic field at point C. [7]

Answer

Distance from each wire to C: $r = \frac{1}{2}\sqrt{(10^2 + 10^2)} = 7.07$ cm	1
$B = \frac{\mu_0 I}{2\pi r}$: $B_1 = 4.53$, $B_2 = 7.07$, $B_3 = 9.05$, $B_4 = 12.4$ ($\times 10^{-5}$ T)	1
B_1 and B_3 are in the same direction: 1.36×10^{-4} T along one diagonal	1
B_2 and B_4 are in the same direction: 1.95×10^{-4} T along the other diagonal	1
The two diagonals are perpendicular	1
$B = \sqrt{((1.36)^2 + (1.95)^2)} \times 10^{-4} = 2.38 \times 10^{-4}$ T	1
Directed towards the I_4–I_3 side, about 10° from the perpendicular towards I_3	1
	7

Explanation: $\tan^{-1}(1.36/1.95) = 35°$ from the C–I_3 diagonal, i.e. 10° from straight down towards I_3.

(iii) State the motion of all wires if the currents I_1, I_2, I_3 and I_4 are of the same magnitude and direction. Give a reason for your answer. [2]

Answer

All four wires move towards C (inwards)	1
Parallel currents in the same direction attract each other	1
	2
Total for question	**15**

Explanation: By symmetry the net force on each wire points to the centre.

20. (a) Explain self-induction in a coil. [2]

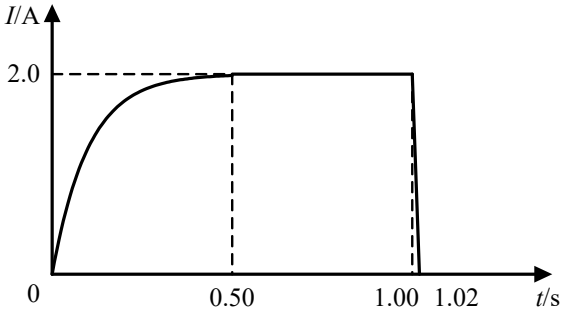
Answer

A changing current in a coil produces a changing magnetic flux that links the coil itself	1
An e.m.f. (back e.m.f.) is induced in the coil that opposes the change in current (Lenz's law)	1
	2

Explanation: $\varepsilon = -L\,\mathrm{d}I/\mathrm{d}t$.

(b) An inductor of self-inductance 2.00 H is connected in series to a battery of e.m.f. 6.00 V and a vibrating switch. A neon lamp which needs a minimum voltage of 100 V to light up is connected

parallel to the inductor. The variation of current I flowing through the inductor with time t is shown in a sketch of the graph below.



The switch is closed at $t = 0$ s and opened at $t = 1.00$ s.

(i) Determine the resistance of the inductor. [2]

Answer

$R = \frac{E}{I_{max}} = \frac{6.00}{2.0}$ 1

$R = 3.0\ \Omega$ 1

2

Explanation: At the steady current there is no back e.m.f.

(ii) Give a reason for the delay in the current to achieve the maximum value after the switch is closed. [1]

Answer

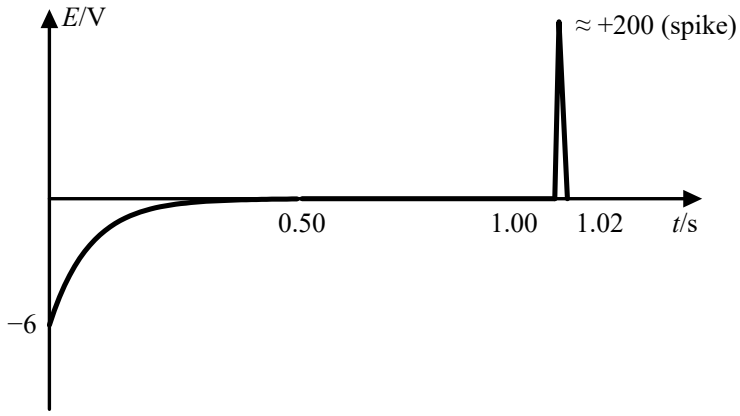
The back e.m.f. induced in the inductor opposes the growth of the current 1

1

Explanation: Time constant $L/R = 0.67$ s.

(iii) Sketch a graph of the variation of induced e.m.f. E in the inductor with time t. [2]

Answer



Back e.m.f. of −6 V at $t = 0$ decaying to zero while the current is steady	1
A large e.m.f. of opposite sign (spike) when the switch opens at 1.00 s	1
	2

Explanation: The induced e.m.f. is proportional to $-\mathrm{d}I/\mathrm{d}t$.

(iv) Based on your graph in (iii), describe how the neon lamp can be lighted up. [2]

Answer

When the switch opens, the current falls to zero very quickly (in 0.02 s)	1
The large induced e.m.f. (about 200 V) across the inductor exceeds 100 V, so the neon lamp lights momentarily	1
	2

Explanation: The 6.0 V battery alone cannot light it.

(v) Calculate the initial rate of change of the current when the switch is closed. [2]

Answer

$\frac{\mathrm{d}I}{\mathrm{d}t} = \frac{E}{L} = \frac{6.00}{2.00}$	1
$= 3.0\ \mathrm{A\ s^{-1}}$	1
	2

Explanation: At $t = 0$ the current is zero, so the whole e.m.f. is across L.

(vi) Calculate the average induced e.m.f. E in the inductor when the switch is opened. [2]

Answer

$E = L\frac{\Delta I}{\Delta t} = (2.00)\frac{2.0}{0.02}$ 1

$E = 200$ V 1

2

Explanation: The current falls from 2.0 A to zero between 1.00 s and 1.02 s.

(vii) Calculate the maximum energy which is stored in the inductor. [2]

Answer

$U = \frac{1}{2}LI^2 = \frac{1}{2}(2.00)(2.0)^2$ 1

$U = 4.0$ J 1

2

Total for question **15**

Explanation: This energy is released when the switch opens.